lab14

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Background

The data from the project came from a knockdown experiment studying the effects of the HOX gene.

Data Import

```
counts <- read.csv("GSE37704_featurecounts.csv", row.names = 1)
metadata <- read.csv("GSE37704_metadata.csv", row.names = 1)

head(counts)
```

```
                 length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
                 SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

```r
metadata
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

Clean up

Checking for ids to align from counts and metadata

```r
new.counts <- counts[2:ncol(counts)]
all(colnames(new.counts) == rownames(metadata))
```

```
[1] TRUE
```

Removing zero count genes

```r
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
new.counts <- new.counts %>%
  filter(rowSums(new.counts) > 0)
```

DESeq Analysis

Imput Object

Loading the `DESeq2` library

```r
library(DESeq2)
```

```
Warning: package 'matrixStats' was built under R version 4.4.3
```

Loading the data into a variable

```r
dds <- DESeqDataSetFromMatrix(countData = new.counts,
                        colData = metadata,
                        design = ~condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

Running DESeq

Running the `DESeq()` function

```r
dds <- DESeq(dds)
```

```
estimating size factors
```

```
estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

Results

Running the results function to veiw the results of the DESeq output.

```
res <- results(dds)
```

Volcano plot

Creating a new volcano plot to view the data

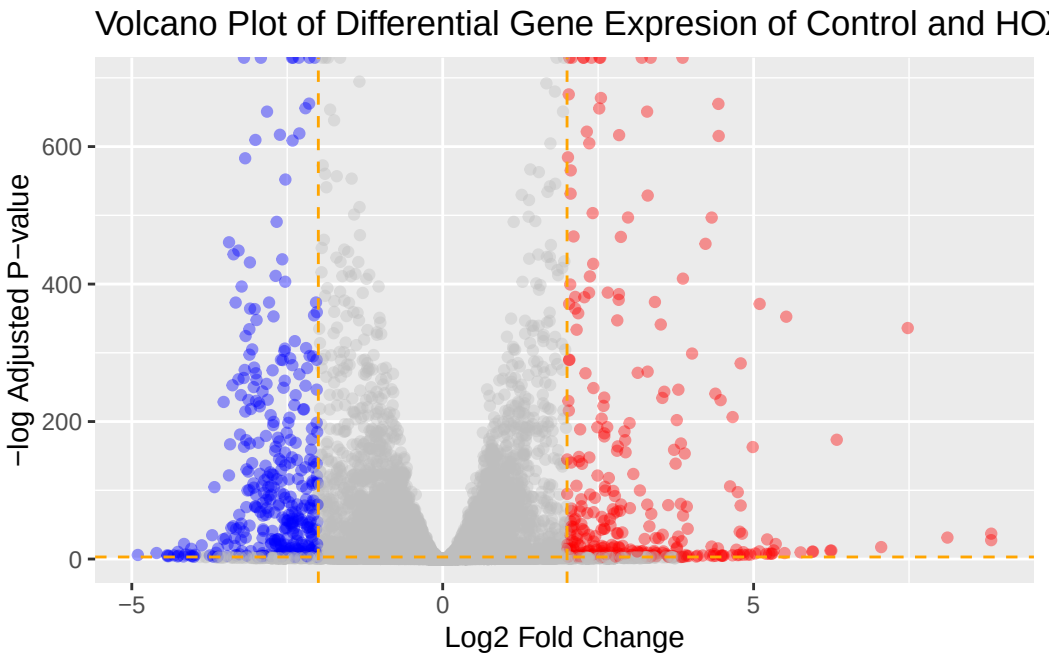
```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
my_cols <- rep("gray", nrow(res))
my_cols[res$log2FoldChange > 2] <- "red"
my_cols[res$log2FoldChange < -2] <- "blue"
my_cols[res$padj >0.05] <- "gray"

ggplot(res, aes(log2FoldChange, -log(padj))) +
  geom_point(color = my_cols, alpha = 0.4) +
  geom_vline(xintercept = c(-2,2), col = "orange", lty = 2) +
  geom_hline(yintercept = -log(0.05), col = "orange", lty = 2) +
  labs(x = "Log2 Fold Change", y = "-log Adjusted P-value", title = "Volcano Plot of Differen
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Adding Annotations

Adding annotation using the `AnnotationDbi` and `org.Hs.eg.db` libraries.

```
library(AnnotationDbi)
```

```
Attaching package: 'AnnotationDbi'

The following object is masked from 'package:dplyr':

    select
```

```
library(org.Hs.eg.db)
```

Using the `mapIDs` function to map the ENSEMBL ids to the gene symbols and ENTRIZ ids

```r
res$symbol <- mapIds(org.Hs.eg.db,
      key = row.names(res),
      keytype = "ENSEMBL",
      column = "SYMBOL")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$entrez <- mapIds(org.Hs.eg.db,
      key = row.names(res),
      keytype = "ENSEMBL",
      column = "ENTREZID")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$name <- mapIds(org.Hs.eg.db,
      key = row.names(res),
      keytype = "ENSEMBL",
      column = "GENENAME")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
head(res)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 9 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9136      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.9798      0.5428105 0.5215598   1.040744 2.97994e-01
                      padj      symbol      entrez                   name
                  <numeric> <character> <character>            <character>
ENSG00000279457 6.86555e-01          NA          NA                     NA
ENSG00000187634 5.15718e-03      SAMD11      148398 sterile alpha motif ..
ENSG00000188976 1.76549e-35       NOC2L       26155 NOC2 like nucleolar ..
```

```
ENSG00000187961 1.13413e-07      KLHL17      339451 kelch like family me..
ENSG00000187583 9.19031e-01     PLEKHN1       84069 pleckstrin homology ..
ENSG00000187642 4.03379e-01       PERM1       84808 PPARGC1 and ESRR ind..
```

Saving results data

```
write.csv(res, "results.csv")
```

Pathway Analysis

Loading the `gage`, `gageData`, and `pathview` packages to highlight gene pathways influenced by HOX.

```
library(gage)
library(gageData)
library(pathview)
```

Preparing data for viewing in pathview

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
       <NA>      148398       26155      339451       84069       84808
 0.17925708  0.42645712 -0.69272046  0.72975561  0.04057653  0.54281049
```

KEGG

Imputing data into the `gage()` function

```
data(kegg.sets.hs)

keggres = gage(foldchanges, gsets = kegg.sets.hs)
head(keggres$less)
```

```
                                                   p.geomean stat.mean
hsa04110 Cell cycle                             8.995727e-06 -4.378644
hsa03030 DNA replication                        9.424076e-05 -3.951803
hsa05130 Pathogenic Escherichia coli infection 1.405864e-04 -3.765330
```

```
hsa03013 RNA transport                          1.246882e-03 -3.059466
hsa03440 Homologous recombination               3.066756e-03 -2.852899
hsa04114 Oocyte meiosis                         3.784520e-03 -2.698128
                                                       p.val       q.val
hsa04110 Cell cycle                             8.995727e-06 0.001889103
hsa03030 DNA replication                        9.424076e-05 0.009841047
hsa05130 Pathogenic Escherichia coli infection 1.405864e-04 0.009841047
hsa03013 RNA transport                          1.246882e-03 0.065461279
hsa03440 Homologous recombination               3.066756e-03 0.128803765
hsa04114 Oocyte meiosis                         3.784520e-03 0.132458191
                                                set.size         exp1
hsa04110 Cell cycle                                  121 8.995727e-06
hsa03030 DNA replication                              36 9.424076e-05
hsa05130 Pathogenic Escherichia coli infection        53 1.405864e-04
hsa03013 RNA transport                               144 1.246882e-03
hsa03440 Homologous recombination                     28 3.066756e-03
hsa04114 Oocyte meiosis                              102 3.784520e-03
```

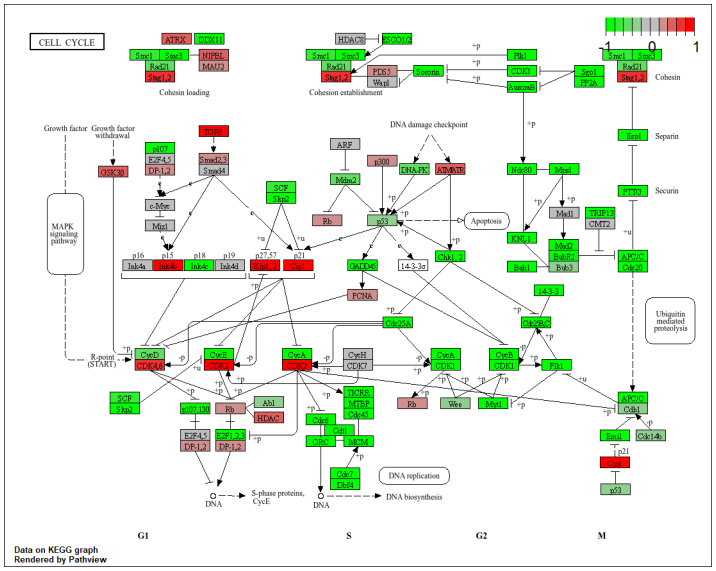
visualizing the pathway with pathview

```
pathview(foldchanges, pathway.id = "hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/eswan/OneDrive/Desktop/College Work/BIMM 143/Lab work/lab

Info: Writing image file hsa04110.pathview.png
```



GO

Imputing data into the `gage()` function

```
data(go.sets.hs)
data(go.subs.hs)

gores <- gage(foldchanges, gsets = go.sets.hs[go.subs.hs$BP])
head(gores$less)
```

```
                                          p.geomean stat.mean        p.val
GO:0048285 organelle fission             1.536227e-15 -8.063910 1.536227e-15
GO:0000280 nuclear division              4.286961e-15 -7.939217 4.286961e-15
GO:0007067 mitosis                       4.286961e-15 -7.939217 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.169934e-14 -7.797496 1.169934e-14
GO:0007059 chromosome segregation        2.028624e-11 -6.878340 2.028624e-11
GO:0000236 mitotic prometaphase          1.729553e-10 -6.695966 1.729553e-10
                                                 q.val set.size         exp1
GO:0048285 organelle fission             5.841698e-12      376 1.536227e-15
GO:0000280 nuclear division              5.841698e-12      352 4.286961e-15
GO:0007067 mitosis                       5.841698e-12      352 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.195672e-11      362 1.169934e-14
GO:0007059 chromosome segregation        1.658603e-08      142 2.028624e-11
GO:0000236 mitotic prometaphase          1.178402e-07       84 1.729553e-10
```

Reactome

For Reactome, we need to prepare our data into a tabular format to download and load into the Reactome website.

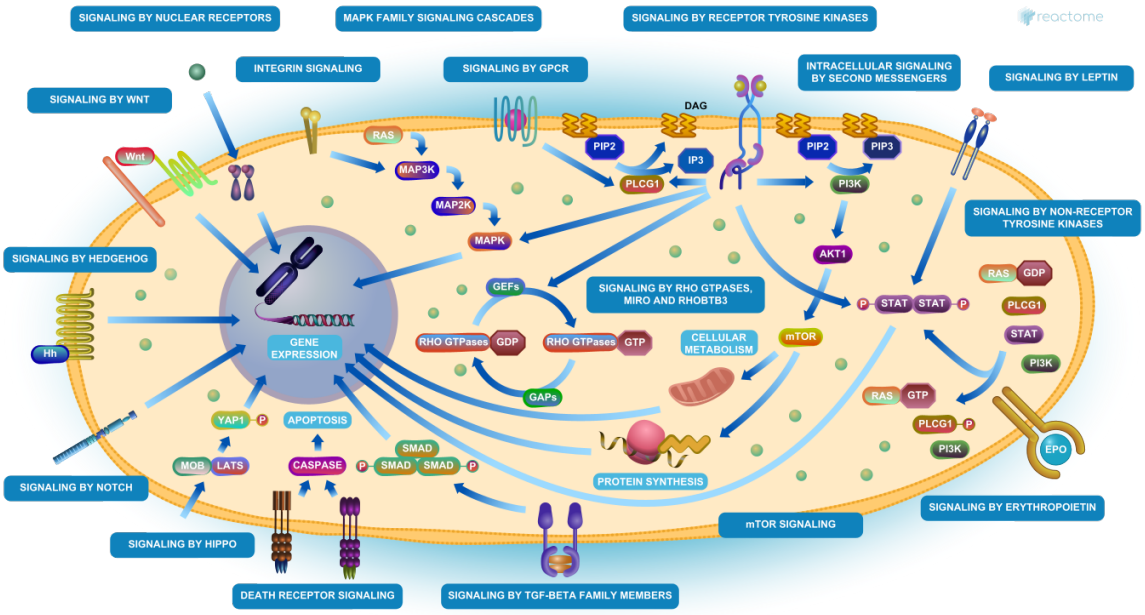
```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"] # filter genes for non NA and
length(sig_genes)
```

```
[1] 8147
```

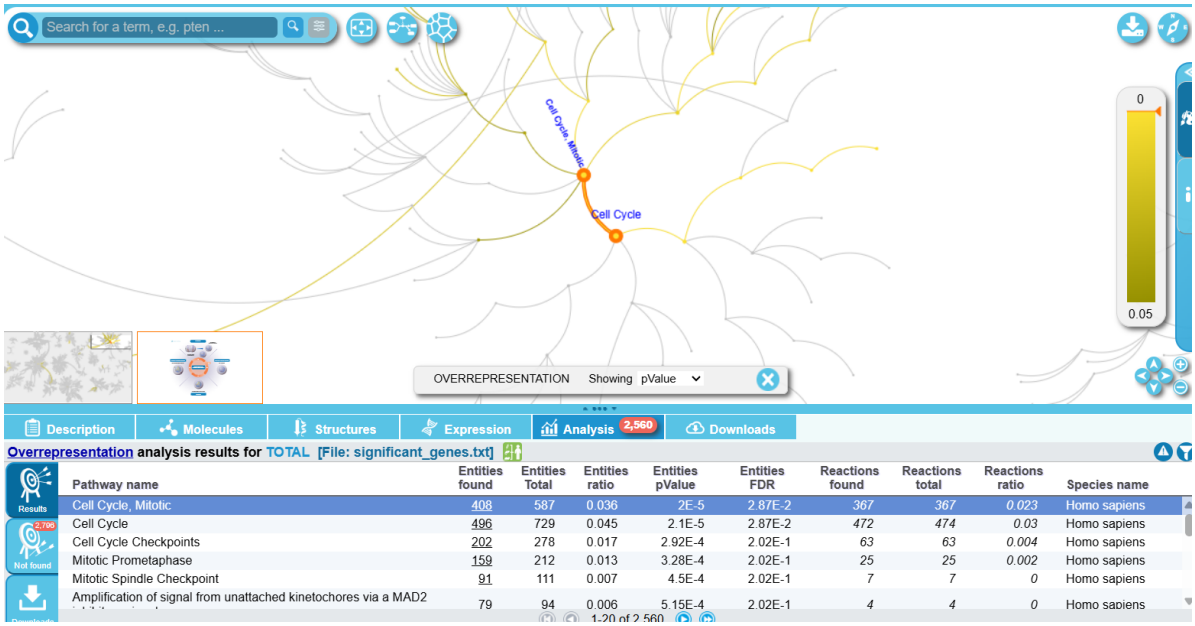
Writing a table to input into Reactome

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

From here we can take the file and analyze it using their online website.



> Q: What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?



The most significant pathway was the mitotic cell cycle pathway which did align with the KEGG results. Differences might result due to the different data sets that KEGG and Reactome use.